

COMMITMENT DAY

Predictable Energy

Strengthening Resilience and Reducing Uncertainty

**DATA CENTRES: CONSTRUCTOR, EPC, ENGINEERING:
AN EFFECTIVE DATA CENTRE SECTOR**

Circulation version
V 1.3



COMMITMENT DAY

Predictable Energy

Strengthening Resilience and Reducing Uncertainty

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Data Centres – Expert panel - Constructor

This session recognises intense pressures on data centre construction and operation under major changes in energy and environmental policy (especially concerning data centres), and builds on the current policy situation about Data Centres.

Relevant regulatory or policy bodies include **Ministry of Energy (NEPC/EPPO, ERC (Min Energy) BOI, NBTC, Min Industry, Data Centre Policy Committee**. The previous session focused on the policy issues. This Session works with what is, what is in development, and the essential ingredients for a sustainable, viable cost effective, data centre sector.

What are the investment economics? Operational economics? What end-to-end cost and efficiency improvements are possible? Optimal energy sources, and water issues. What are the various cost elements in tariffs (eg UGT2, current pilot Direct PPA 2,000 MW of DCs).

What are some attractive world practices? What future innovations are in the pipeline?

PREDICTABLE ENERGY includes availability, cost and confidence in transformation to higher renewables demand, and a decentralised sector away from a traditional monopoly. Is there a positive impact on the economics of data centres?

Data Centres Expert Panel - Constructor

Mr Chalai van Kampen Head of Cost Management (Data Centres, Industrial & Logistics) JLL

Mr Matthew Silvester Director at Meinhardt Thailand

Mr Ibrahim Kocagoz, Engineering Solutions Director, PCS Thailand

Mr Lucas Meier Program Manager at the German–Southeast Asian Center of Excellence for Public Policy and Good Governance (CPG)

Moderator: Mr Bob Fox – Chair Digital Economy/ICT JFCCT and JFCCT Policy Co-Ordinator



EXPERT PANEL CONSTRUCTOR, EPC, ENGINEERING



Matthew Silvester

Director
Meinhardt (Thailand) Ltd.



Chailai van Kampen

Head of Cost Management (Data
Centres, Industrial & Logistics)
JLL



Ibrahim Kocagoz

Director - Engineering Solutions
PCS Thailand



Lucas Meier

Program Manager
The German–Southeast Asian
Center of Excellence
for Public Policy and Good Governance
(CPG)



Moderator Bob Fox

Chair, Digital Economy /ICT
and Policy Co-Ordinator
Joint Foreign Chambers
of Commerce in Thailand (JFCCT)

Question themes – Constructor panel

THEME ONE: Economics – General and Specific

THEME TWO: The new energy regime means what?

THEME THREE: A future industry, including a decentralized energy sector



Data Centres end-to-end issues – JFCCT Digital Economy/ICT consistently since Dec 2024

DATA

Data Governance / collection

Analysis / Processing

PROCESSING

Photonics, chips

LLMs vs SLMs

AI model consumption

RESOURCES

Energy / Renewable Energy PUE

Special RE tariffs

Floating solar

Water usage WUE

Possible national rationing / priority

CONSTRUCTION

Wood

Under water

NATIONAL POLICY

Over promotes DCs ?

NBTC – cat 3 licence?

BOI conditions

Data Localisation

Location and proximities/ zoning

NEPC

DC Policy Committee

REGIONAL

ASEAN Power Grid

ASEAN co-operation

Other regional practice – conditions

APDCA – Sustainable Digital Infrastructure Accord

[\(SDIA\);](#)

MODE

Terrestrial

Orbital

Ocean

Digital Economy policy – the regional hub will not be decided by the economy with most⁷ Data Centre space.

Data Centres – Expert panel - Constructor

Following four slides –

Chalalai van Kampen Head of Cost Management (Data Centres, Industrial & Logistics) JLL



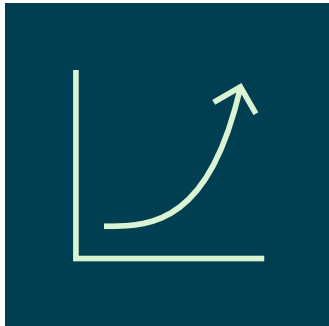
Commitment Day

**Navigating AI demand, power
constraints and global opportunities**

JLL Global Research
2026



Hyperscale data centres are growing their presence, Country-wide capacity might witness approximately 1,500 MW of known future supply within 2030



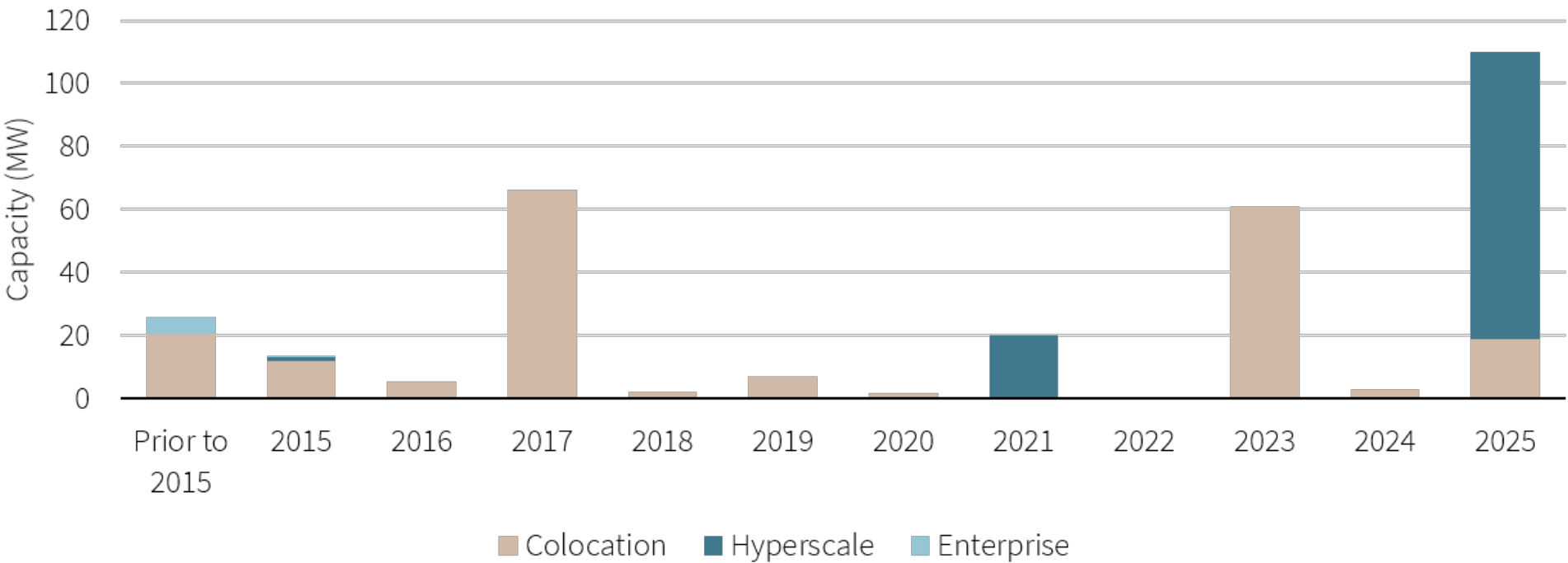
The previous trend of data centre was colocation. Now, it is time for hyperscale data centres.

Prior to 2025, colocation data centres have dominated the market. However, the trend has shifted towards hyperscalers. Eastern Economic Corridor (EEC) is one of the prime location for large-scale infrastructure, while locations near Bangkok’s CBA are in high demand for colocations.

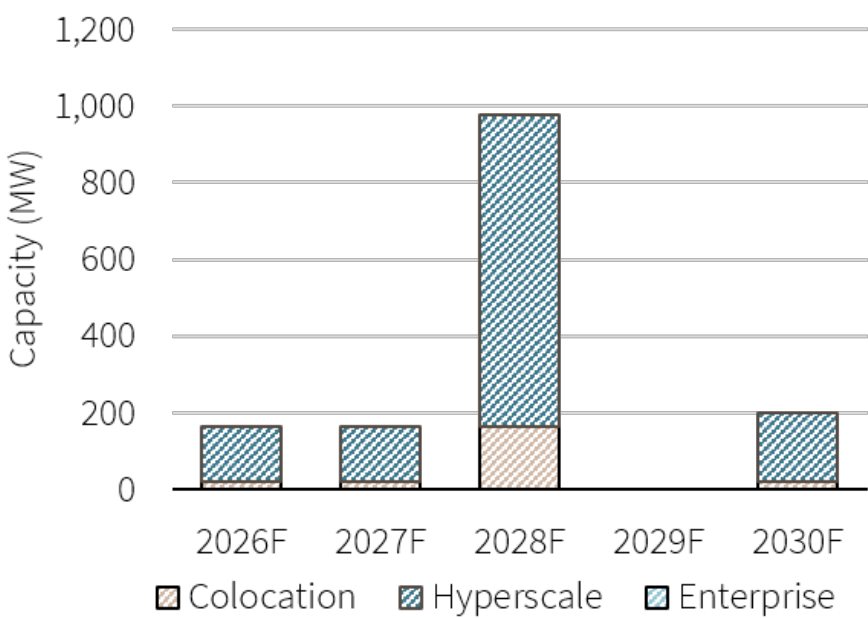
Future pipeline looks promising for Thailand.

Thailand has become a hotspot for global data centre operators in recent years, with intensive government promotion by the Board of Investment (BOI). The majority of future players will be hyperscalers. Live capacity will likely to take place in 2026-2027.

Historical Supply, 2015-2025



Future Supply, 2026-2030F



Source: JLL Research

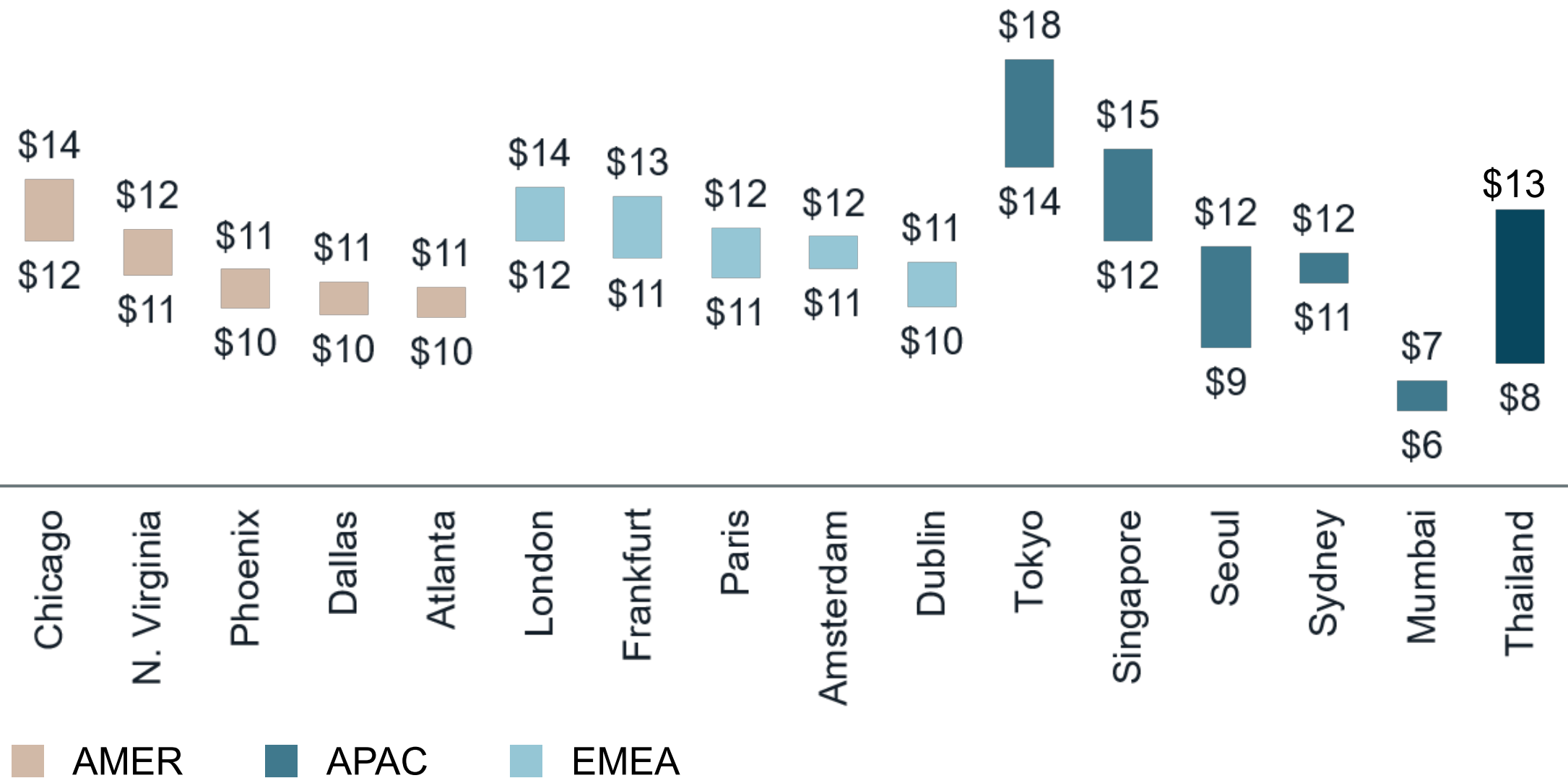
Data center construction costs have been increasing 7% annually



Between 2020 and 2025, the average cost increased from \$7.7 to \$10.7 million per MW, representing an average 7% increase in year.

The cost per MW in 2026 is projected to reach \$11.3 million.

2026 average data center construction costs (\$ millions per MW)



Source: JLL Research

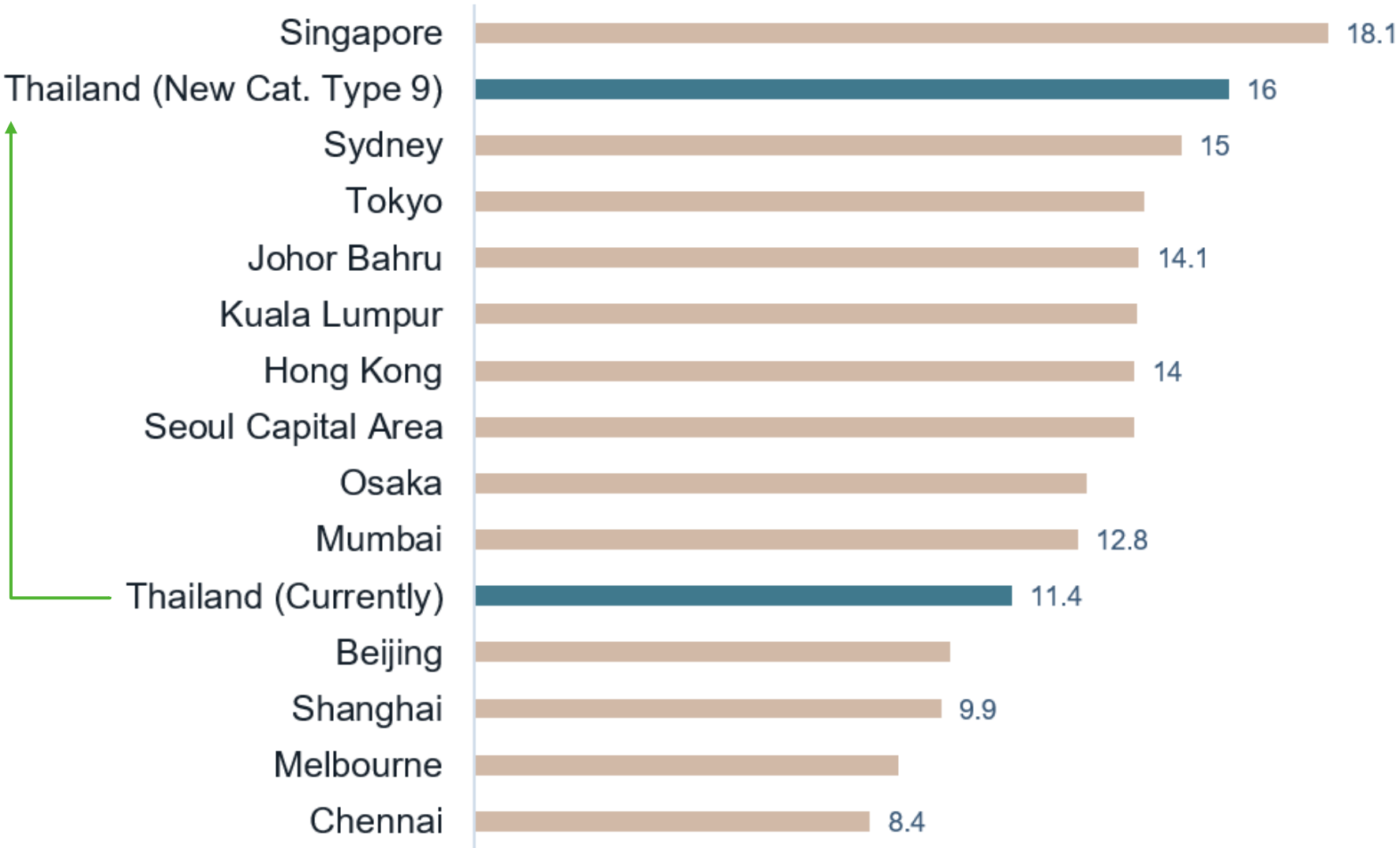
75% of development is concentrated in low-cost electricity markets



Power availability will influence industry growth as operators address constraints through behind-the-meter power options.

Solar energy is expected to be predominant renewable power option in APAC.

Asia Pacific colocation electricity rates (US cents/kWh)



Source: JLL Research

Data Centres – Expert panel - Constructor

Following three slides –

Matthew Silvester Director at Meinhardt Thailand

DELIVERY BARRIERS AND THE POLICY FIX

- Grid connection, land and water are the hard limits
- Clearer permitting rules, specific for DC
- Prescriptive requirements for BOI
- Clarity on process and engagement with policy makers



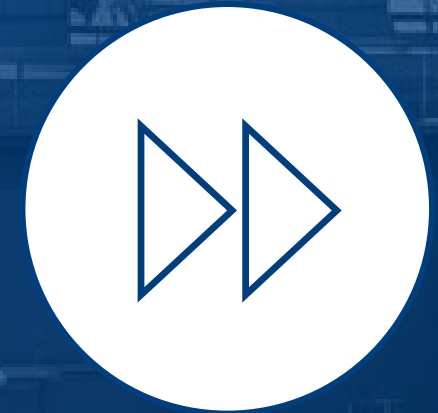
IS POWER REALLY THE CONSTRAINT?

- Capacity is adequate in aggregate
- The issue is where and when power lands
- Transparency on connection timeline
- Grid planning and PPA matter most



SPEED TO MARKET AND THE WIDER CAMPUS

- Expedited permitting
- Off-site prefabrication and modular plant
- Supply chain & labour constraints
- Energization



Data Centres – Expert panel - Constructor

Following four slides –

Ibrahim Kocagoz, Engineering Solutions Director, PCS Thailand

pos | Engineering

Smart Operations and Energy Efficiency.

Ibrahim Kocagoz

Engineering Solutions Director

2026



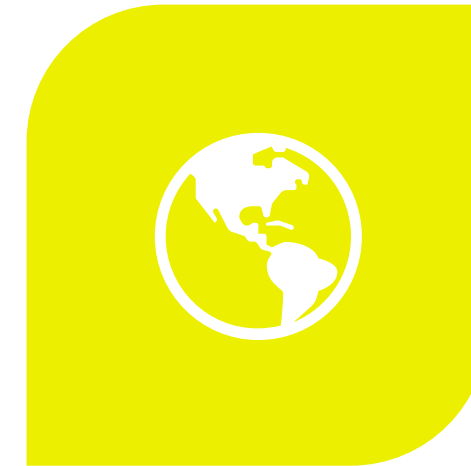
Challenges



RESOURCES (ELECTRICITY,
WATER, SKILLED PEOPLE)



HIGH DEMAND



GLOBAL CRISES IMPACTING
LOGISTICS AND COST

Bringing people & Technology together

Virtual Technician / Smart Glasses

Benefits

- Hands free online camera recording, while reviewing the drawing on the small screen
- Voice command (take photo, open drawing, connect to a meeting, etc.), so technician can use both hand while talking, recording video, talking to Command Center
- Efficient O&M service
- A great tool for Remote operation and Centralized Control
- Record the real time data all the time and store it.



Overview of PCS Thailand



59 years



6,000+ Sites



38,000+
colleagues



Multiple Training
Academy Centers Across
Thailand



16 branches across
Thailand



Industry leaders
in Facility Services

Our Accreditation



ISO 9001
Quality
Management
Systems
CERTIFIED

ISO 14001
Environmental
Management
CERTIFIED

ISO 45001
Occupational
Health and Safety
Management
CERTIFIED

ISO 41001
Facilities
Management
CERTIFIED



6 Certified Security Training Centre Nationwide

Bangkok

Chiang Mai

Chonburi

Khon Kaen

Phuket

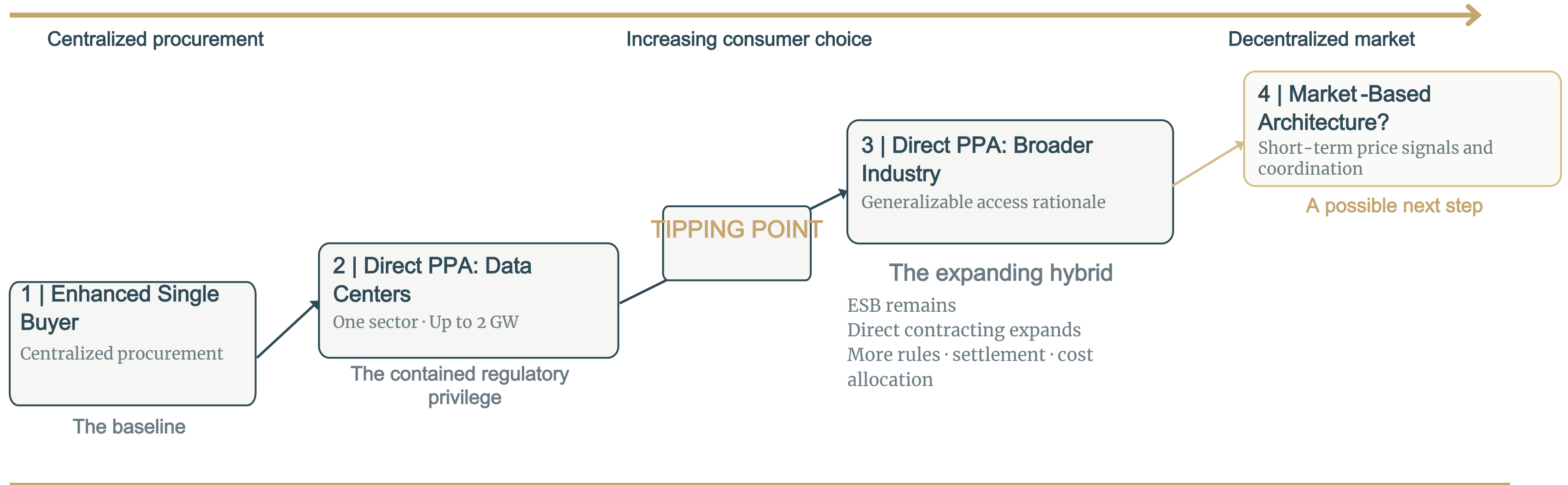
Songkhla

Data Centres – Expert panel - Constructor

Following one slide –

Mr Lucas Meier Program Manager
German–Southeast Asian Center of Excellence for Public Policy and Good Governance (CPG)

Direct PPA and the Evolution of Thailand's Electricity Market



Hybrid as the endpoint —or Direct PPA as a stepping stone?

The market path is already part of Thailand's policy debate

Question themes – Constructor panel

THEME ONE: Economics – General and Specific

THEME TWO: The new energy regime means what?

THEME THREE: A future industry, including a decentralized energy sector

THEME ONE: Economics – General and Specific

1. Setting the scene. We have looked at current status of energy policy changes and development, at a high level, what impact will this have on Data Centre economics? Would allocation and rationing be inevitable or can that be addressed by economics?
2. The proposed Direct PPA pilot is not closed, but how is it changing the way you model capex/opex?
3. From a cost management perspective, what's the single biggest source of uncertainty in pricing a Thai data centre project today?
4. Different renewable sources – best guess on viability, on cost? (solar, wind, SMR, hydro, other, and consider BSS)

THEME TWO: The new energy regime means what?

5. We seem to be moving away slightly from a policy aimed at building the largest Data Centre floor space in S E Asia on the basis that somehow that will mean being a digital hub. Does that seem right?

6. With data centres now facing a separate, higher-priced electricity class and a guarantee requirement just to apply, plus other changes, what does that do to project economics?"

7. Water and energy get most of the policy attention, are there other costs quietly escalating that people aren't talking about? And with cost normalization over time as the industry has more experience with these costs, where the cost drivers are better understood, what is essential, what is not?

THEME THREE: A future industry, including a decentralized energy sector

8 Thailand's policy direction – is it now more about attracting investment, or about proving a viable sector with various constraints?

9. Direct PPAs may be short to medium term exceptions to a decentralised, market based but regulated price and availability for energy for many uses, what is a longer term vision for a decentralised industry?



PREDICTABLE ENERGY includes

- availability
- cost
- confidence in transformation to higher renewables demand, and a decentralised sector away from a traditional monopoly